

Predicting When Collection Statistics Help Top- k Selection: From Retrieval to Attention and Expert Routing

Team ANLP Doomsday

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Abstract

Learned sparse retrieval, block-sparse attention and mixture-of-experts routing all do the same thing. Each scores a query against a set of learned non-negative features by inner product and keeps the top- k , and none of them uses how often a feature fires across the collection, which is the statistic BM25 is built on. Whether that omission costs anything is unsettled even in retrieval, where replacing BM25 with a plain inner product over the same vocabulary drops nDCG@10 from 0.582 to 0.381 on sparse-autoencoder features and raises it from 0.455 to 0.568 on SPLADE. The published account is that BM25 helps while a vocabulary stays Zipfian, but that has only been observed one vocabulary at a time and never turned into a prediction. We build eight concept vocabularies from one frozen encoder, score each with a BM25-style function whose components toggle independently, and regress the resulting margin on six cheap distributional measurements. The fitted predictor then goes, without refitting, to KV-page selection and expert routing. Whether it holds there is open. Attention sinks fire constantly yet cannot be removed, and the rarest experts in a mixture-of-experts are the most critical, so frequency may signal importance differently inside a model than across a corpus.

1 The Same Computation in Three Places

A learned sparse retriever scores a query-document pair as $\langle u_q, u_d \rangle$ over a concept vocabulary. A block-sparse attention kernel estimates how critical KV page i is for query q_t as $r_{t,i} = \langle q_t, c_i \rangle$ over page summaries and attends to the top- k pages only (Tang et al., 2024). A mixture-of-experts router gates on $g_e = \langle h_t, w_e \rangle$ and sends each token to its top- k experts. These come from three literatures and perform one computation, an inner product against a learned non-negative basis, then a hard top- k .

None of them asks how often a feature wins.

BM25 has corrected exactly that for decades, discounting features that fire everywhere through inverse document frequency, saturating repeated evidence, and normalising for length (Robertson and Zaragoza, 2009). None of those ingredients needs words. Each needs only a representation with non-negative coordinates, a meaningful per-dimension collection frequency, and a sensible notion of magnitude, and learned sparse features supply all three. Fragments of the correction keep reappearing unrecognised. The per-expert load bias of loss-free balancing is an additive frequency discount with no saturation or length term (Wang et al., 2024), and streaming inference hard-codes an exception for the highest-frequency keys (Xiao et al., 2024). The need is felt case by case, but no general account says when the correction helps.

2 What Is Already Known

Clavié et al. (2026) show that sparse-autoencoder features taken from a frozen dense retriever have roughly Zipfian collection statistics and are directly BM25-scorable. Their Appendix C carries the sharpest evidence. On one suite BM25 reaches 0.582 nDCG@10 against 0.381 for the inner product on SAE latent terms, and 0.455 against 0.568 on SPLADE-v3, so the two scorers swap places when only the vocabulary changes. Han et al. (2026) run the same comparison on image features and find the two within about two points of R@1 once top- k filtering is applied, which they attribute to top- k already suppressing what IDF would down-weight. Two further papers build SAE-based sparse retrievers and score them by inner product without ever running the comparison (Formal et al., 2026; Zong et al., 2026).

Both papers that did compare propose the same mechanism, that BM25 helps while a vocabulary keeps its Zipfian structure. That explanation is already in print and we are not claiming it. It has

only been seen one vocabulary at a time, never turned into a prediction, and never tested outside retrieval. CL-SR (Park et al., 2025), the nearest published system, indexes SAE-decomposed embeddings as concept units, whereas we leave representation learning alone and study the scorer.

3 Why the Transfer Might Fail

Inside the transformer, how often a feature fires and how much it matters come apart in both directions. Attention sinks are the most frequently attended keys in a cache, which by IDF logic makes them the least informative, yet deleting them collapses generation (Xiao et al., 2024). Su et al. (2026) report the mirror case, where removing three experts out of 6,144 badly degrades Qwen3-30B-A3B, so the rarest experts are the most critical. IDF predicts that second case, but would also promote thousands of merely rare experts. A recent KV-cache method ranks entries by expected future attention and prunes the low scorers (Devoto et al., 2025), the opposite ordering to IDF, and it works. Either the map from frequency to importance inverts between a corpus and a model, or two quantities have been conflated, how much a feature contributes to the output and how much it separates one query from another. Distinguishing them is what the transfer experiment is for.

4 Plan

We hold representation learning fixed and vary the basis and the scorer independently. Each setting gets one scoring function in which soft IDF, saturation, length normalisation and top- k sparsification are independent switches, so the plain inner product is the all-off case and BM25 the all-on case, and any gain is attributable to a mechanism rather than a bundle.

Bases and data. Eight bases come from one frozen encoder and span near-dense to strongly lexical, namely split-rectified identity, random projection, PCA, k -means, product and residual quantisation, a Top- K SAE, and SPLADE (Formal et al., 2021). SPLADE matters because the reversal we are explaining is SPLADE against SAE. Running every basis at pooled and token-level granularity gives sixteen representations, making granularity a variable rather than a confound. We tune only on MS MARCO, add LIMIT (Weller et al., 2025) because it separates lexical from semantic signal by construction, and transfer zero-shot to a BEIR

subset and TREC-DL 19/20, reproducing the Appendix C reversal first. Encoding MS MARCO once dominates the cost, since every basis is then a linear map over cached embeddings.

Languages. Collection frequency belongs to a corpus and a language rather than to a model, so a predictor built on it should be tested where the distribution moves. No prior SAE retriever covers an Indic language, and the multilingual evaluation of Zong et al. (2026) spans six MIRACL languages, none Indian. We repeat the grid on the Hindi, Bengali and Telugu portions of MIRACL (Zhang et al., 2023) and on Hindi-English code-mixed queries, where transliteration and mixing flatten the document-frequency curve. Corpus-level measures of mixing already predict downstream behaviour there (Kodali et al., 2022), and our diagnostics are instruments of the same kind.

Predictor. One forward pass per basis gives active fraction, the Zipf exponent of the selection-frequency curve, Gini of activation mass, head mass, IDF variance, and the score mass held by the top 1% of features. Han et al. (2026) report several descriptively but never predict with them. We regress the BM25-minus-inner-product margin on the six and validate by holding out whole basis families, so the predictor is never scored on a construction method it has seen.

Transfer. The fitted predictor then goes unchanged to the two transformer settings. For KV selection we take Quest page criticality (Tang et al., 2024) and weight it by $\text{idf}(\text{sf}_i)$, where sf_i is the fraction of prefill queries for which page i enters the top- k ; the saturation and length switches carry over to the page score and its norm. The correction needs no training, and whether sinks are exempted from it is an ablation rather than an assumption. For routing we apply the same weighting to router logits at inference on OLMoE-1B-7B (Muenighoff et al., 2025), against loss-free balancing as the training-time special case.

What would count as a result. Confirming that BM25 helps on Zipfian vocabularies restates what both comparison papers predict, so the useful outcomes are the awkward ones. A predicted gain for KV selection where the measured effect is negative would separate contribution to the output from discrimination between queries. Diagnostics promising a gain where none appears would falsify the Zipfian account.

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